Two lemmas for route B

(canonical constants and an a priori bound for quasi-solutions of the rung system)

Notes to `rvm_zeta_pi_mertens_v18`

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Conventions. All references are to v18. τ_k is the midpoint clock, $n(t) := \#\{k : \tau_k \le t\}$, $\mathcal{N}_{\mathrm{RVM}}(t) = \frac{t}{2\pi}\log\frac{t}{2\pi e} + \frac{7}{8}$, $G(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)$, $q(s) = s(s-1)$, and for a nondecreasing real list $\boldsymbol{\kappa} = (\kappa_k)_{k\ge1}$ with $\kappa_1 \ge \tau_0^+$ and $|\kappa_k - \tau_k|$ bounded,

$$P_{\boldsymbol{\kappa}}(s) = \prod_{k\ge1}\Big(1 + \frac{q(s)}{\frac14 + \kappa_k^2}\Big), \qquad \zeta_{\boldsymbol{\kappa}} = \frac{P_{\boldsymbol{\kappa}}}{G}, \qquad N_{\boldsymbol{\kappa}}(t) = \#\{k : \kappa_k \le t\},$$

as in (eq:list-objects). $P_{\boldsymbol{\kappa}}$ is entire, real on the real axis, symmetric under $s \mapsto 1-s$, positive on $(1,\infty)$, and its zeros are exactly $\frac12 \pm i\kappa_k$ (Theorem thm:path in its list form). The rung system is (eq:rung-system),

$$\sum_{k\ge1}\cos(\kappa_k u) + \sum_{n\ge2}\mathsf{a}_n\,\delta(u - \log n) = \mathsf{k}(u) \qquad \text{in } \mathcal{D}'(0,\infty), \tag{1}$$

with k as in (eq:transport-defs). Nothing below uses the ordinates of ζ.

Lemma 1: the constants of route B are canonical

Lemma 1 (Canonical constants)**.** *Let $(\boldsymbol{\kappa}, (\mathsf{a}_n))$ be a solution of (1) with $\boldsymbol{\kappa} \in \mathscr{O}$. Then*

$$|\kappa_k - \tau_k| \le C_\eta \quad (k \ge 1), \qquad 0 \le \mathsf{a}_n \le \frac{1}{e} \quad (n \ge 2),$$

where C_η is the constant of Lemma lem:bounded-displacement. Consequently, if bounded finite sections of (1) are solved under the constraints $|\kappa_k - \tau_k| \le C$ and $|\mathsf{a}_n| \le M$, nothing is lost by taking

$$C = C_\eta, \qquad M = \tfrac{1}{e},$$

and route B is the single parameter-free statement $E_m(C_\eta, \frac1e) \to 0$.

Proof. By Theorem thm:free-masses the solution is unique, $\boldsymbol{\kappa} = (\gamma_k)$ and $\mathsf{a}_n = \Lambda(n)/(2\sqrt{n})$. The first bound is Lemma lem:bounded-displacement. For the second, $\mathsf{a}_n \le \frac{\log n}{2\sqrt{n}}$, and $x \mapsto \frac{\log x}{2\sqrt{x}}$ has its maximum on $[1,\infty)$ at $x = e^2$, where it equals $1/e$. □

Remark 1. *Theorem thm:free-masses is stated for $\boldsymbol{\kappa} \in \mathscr{O}$, and $\mathscr{O}$ carries, besides the tube $|\kappa_k - \tau_k| \le C_\eta$, the primitive-count condition $|\int_{\tau_0}^X (N_{\boldsymbol{\kappa}} - \mathcal{N}_{\mathrm{RVM}})| \le C\log X$ of (eq:list-space). A list obtained as a coordinatewise limit of finite sections inherits the tube and the ordering, not that condition. Before Lemma 1 is applied to such a limit, Theorem thm:free-masses must be checked for exact solutions of (1) that satisfy the tube only. For an exact solution the half-plane boundedness of $\zeta_{\boldsymbol{\kappa}}$, which is what the primitive-count condition is used for, follows directly from (1) (Lemma 2, Step 1 with $E \equiv 0$), so the proof should go through; this has to be written.*

Lemma 2: an a priori bound for quasi-solutions

Throughout, $\sigma_0 := 2$, $r := \sigma_0 - \tfrac12 = \tfrac32$, $R := 2$.

Section data. A *section list* is a nondecreasing real list with $\kappa_1 \ge \tau_0^+$, $|\kappa_k - \tau_k| \le C$ for all k and $\kappa_k = \tau_k$ for $k > K$. Its *masses* are real numbers a_n, $2 \le n \le N$, with $|\mathsf{a}_n| \le M$; put $\mathsf{a}_n = 0$ for $n > N$. The *Laplace residual* of $(\boldsymbol{\kappa}, (\mathsf{a}_n))$ is the function

$$\widehat{\mathsf{R}}(s) := \tfrac12 \frac{P_{\boldsymbol{\kappa}}'}{P_{\boldsymbol{\kappa}}}(s) - \tfrac12 \frac{G'}{G}(s) + \sum_{n=2}^{N} \mathsf{a}_n\, n^{\frac12 - s}, \qquad \Re s > 1, \tag{2}$$

which by the distributional explicit formula (Theorem thm:distributional-EF, list form) is the Laplace transform $\int_0^\infty e^{-(s-\frac12)u} \mathsf{R}(u)\, du$ of the residual distribution $\mathsf{R} := \sum_k \cos(\kappa_k u) + \sum_n \mathsf{a}_n \delta(u - \log n) - \mathsf{k}(u)$ of (1); (2) is taken as the definition. The *accumulated residual* is

$$E(T) := \int_{-T}^{T} \bigl|\widehat{\mathsf{R}}(\sigma_0 + iv)\bigr|\, dv, \qquad \varepsilon := \sup_{T \ge 2} \frac{E(T)}{\log T}. \tag{3}$$

Finally $\Lambda_0 := 2\sum_{n\ge2} n^{\frac12 - \sigma_0}/\log n < \infty$ and $c_{\mathrm{RVM}} := \sup_{\Re s \ge \sigma_0} |\log \zeta_{\mathrm{RVM}}(s)|$, finite by Theorem thm:halfplane-RVM.

Lemma 2 (A priori bound for quasi-solutions)**.** *There are absolute constants A_1, A_2, A_3, T_0 such that, for every section list with masses as above and every $T \ge T_0$ with $T \notin \{\kappa_k\}$,*

$$\bigl|N_{\boldsymbol{\kappa}}(T) - n(T)\bigr| \le A_1 \log T + A_2\bigl(1 + c_{\mathrm{RVM}} + M\Lambda_0 + 10^{-2}C\bigr) + A_3\, E(T + R), \tag{4}$$

and consequently, for every k with $\tau_k \ge T_0 + C + R$,

$$|\kappa_k - \tau_k| \le 2\pi\bigl(A_1 + A_3\,\varepsilon\bigr) + \frac{2\pi\bigl(A_2(1 + c_{\mathrm{RVM}} + M\Lambda_0 + 10^{-2}C) + A_3\varepsilon \log(1 + C + R) + 1\bigr)}{\log(\tau_k/2\pi)}\,(1 + o(1)). \tag{5}$$

In particular, with $C^* := 2\pi A_1$: if the section is solved under the constraints $|\kappa_k - \tau_k| \le C$, $|\mathsf{a}_n| \le M$, and its accumulated residual satisfies $\varepsilon \le \varepsilon_0$, then for all k beyond a height depending only on C, M, ε_0,

$$|\kappa_k - \tau_k| \le C^* + 2\pi A_3\, \varepsilon_0 + o(1),$$

so that the tube constraint is not active for $C > C^* + 2\pi A_3 \varepsilon_0 + 1$. The constants do not depend on K, N, C, M or on the section.

Proof. Step 1: the Dirichlet factor on the half-plane. Put

$$\Lambda_{\mathsf{a}}(s) := \sum_{n=2}^{N} \frac{2\mathsf{a}_n}{\log n}\, n^{\frac12 - s}, \qquad Y(s) := \zeta_{\boldsymbol{\kappa}}(s)\, e^{-\Lambda_{\mathsf{a}}(s)}.$$

For $\Re s \ge \sigma_0$, $|\Lambda_{\mathsf{a}}(s)| \le M\Lambda_0$ (this is where $\sigma_0 > \tfrac32$ is needed: the masses are bounded, not decaying). Since $P_{\boldsymbol{\kappa}}$ has no zeros off $\Re s = \tfrac12$ and G has none in $\Re s > 1$, Y is holomorphic and zero-free in $\Re s > 1$, so $\log Y$ has a holomorphic branch there, real on $(1, \infty)$. Differentiating, by (2),

$$(\log Y)'(s) = 2\,\widehat{\mathsf{R}}(s) \qquad (\Re s > 1). \tag{6}$$

At the real point, $\log Y(\sigma_0) = \log \zeta_{\mathrm{RVM}}(\sigma_0) + \sum_{k\le K} \log \frac{(\frac14+\kappa_k^2+q)(\frac14+\tau_k^2)}{(\frac14+\kappa_k^2)(\frac14+\tau_k^2+q)} - \Lambda_{\mathbf{a}}(\sigma_0)$ with $q = q(\sigma_0) = 2$. Each logarithm has modulus at most $\frac{2q\,|\kappa_k-\tau_k|}{\tau_k^3}(1+o(1)) \le \frac{4C}{\tau_k^3}(1+o(1))$, and $\sum_k \tau_k^{-3} < 2.5\cdot 10^{-3}$ (the first terms are $14.5^{-3} + 20.7^{-3} + 25.5^{-3} + \cdots$). Hence

$$|\log Y(\sigma_0)| \le c_{\mathrm{RVM}} + M\Lambda_0 + 10^{-2}C =: B_0.$$

Integrating (6) along the vertical line,

$$|\log Y(\sigma_0 + it)| \le B_0 + 2E(|t|) \qquad (t \in \mathbb{R}). \tag{7}$$

Step 2: Phragmén–Lindelöf in the strip. Let $W(s) := \log|P_{\boldsymbol{\kappa}}(s)| - \log|G(s)| = \log|Y(s)| + \Re\Lambda_{\mathbf{a}}(s)$ in $\Re s > 1$. On the strip $\Sigma := \{1-\sigma_0 \le \Re s \le \sigma_0,\ t \ge 1\}$ the function W is subharmonic: $\log|P_{\boldsymbol{\kappa}}|$ is subharmonic and $\log|G|$ is harmonic there (G has no zeros or poles with $t \ge 1$). On the right side $\Re s = \sigma_0$, by (7) and $E(t) \le \varepsilon \log t$,

$$W(\sigma_0 + it) \le B_0 + M\Lambda_0 + 2\varepsilon \log t.$$

On the left side, the symmetry $P_{\boldsymbol{\kappa}}(1-s) = P_{\boldsymbol{\kappa}}(s)$ gives $W(1-s) = W(s) + \log|G(s)/G(1-s)|$, and $\log|G(\sigma_0+it)/G(1-\sigma_0-it)| = (\sigma_0 - \frac12)\log\frac{t}{2\pi} + O(1/t)$ by Stirling; hence

$$W(1-\sigma_0+it) \le B_0 + M\Lambda_0 + 2\varepsilon\log t + (\sigma_0-\tfrac12)\log\tfrac{t}{2\pi} + c_1.$$

On the bottom side $t = 1$, W is bounded by a constant depending on C only through B_0 (finitely many factors). $W(s) \le c|s|\log|s|$ in Σ because $P_{\boldsymbol{\kappa}}$ is entire of order one with $N_{\boldsymbol{\kappa}}(t) \ll t\log t$. The function $W(s) - (\sigma_0 - \frac12 + 2\varepsilon)\log|s-\frac12|$ is subharmonic in Σ, bounded above by $B_0 + M\Lambda_0 + c_2$ on $\partial\Sigma$, and of growth $o(e^{\pi t/(2\sigma_0-1)})$; the Phragmén–Lindelöf principle for the half-strip gives

$$W(s) \le B_0 + M\Lambda_0 + c_2 + (\sigma_0 - \tfrac12 + 2\varepsilon)\log(2+|t|) \qquad (s \in \Sigma). \tag{8}$$

For $\Re s > \sigma_0$ the same bound holds directly from (7) and Step 1.

Step 3: Backlund's lemma on the segment at height T. Fix $T \ge T_0$, $T \notin \{\kappa_k\}$, and let $\alpha \in \mathbb{R}$ be such that $e^{i\alpha}P_{\boldsymbol{\kappa}}(\sigma_0+iT) > 0$. Put $F(z) := \Re\big(e^{i\alpha}P_{\boldsymbol{\kappa}}(z+iT)\big)$ for real z, which is the restriction to the real axis of the entire function $\frac12\big(e^{i\alpha}P_{\boldsymbol{\kappa}}(z+iT) + e^{-i\alpha}P_{\boldsymbol{\kappa}}(\bar z - iT)^{-}\big)$, where the last term is $e^{-i\alpha}\overline{P_{\boldsymbol{\kappa}}(\bar z + iT)}$ extended analytically; it is entire because $P_{\boldsymbol{\kappa}}$ is real on the real axis. The variation of $\arg P_{\boldsymbol{\kappa}}$ along the segment $[\frac12 + iT, \sigma_0 + iT]$ is at most $\pi(\nu+1)$, where ν is the number of zeros of F on $[\frac12, \sigma_0]$. By Jensen's formula on the disc $|z-\sigma_0| \le R$ (which contains $[\frac12,\sigma_0]$, of radius $r < R$ about σ_0),

$$\nu \log\frac{R}{r} \le \max_{|z-\sigma_0|=R} \log|F(z)| - \log|F(\sigma_0)| \le \max_{|s-(\sigma_0+iT)|\le R} \log|P_{\boldsymbol{\kappa}}(s)| - \log|P_{\boldsymbol{\kappa}}(\sigma_0+iT)|,$$

using $|F(z)| \le \max\{|P_{\boldsymbol{\kappa}}(z+iT)|, |P_{\boldsymbol{\kappa}}(\bar z + iT)|\}$ and $|F(\sigma_0)| = |P_{\boldsymbol{\kappa}}(\sigma_0+iT)|$. On the disc, by $\log|P_{\boldsymbol{\kappa}}| = W + \log|G|$, (8) and the symmetry of W (the disc meets $\Re s < 1-\sigma_0$ only if $R > 2\sigma_0 - 1$, which is not the case for $R = 2$, $\sigma_0 = 2$),

$$\max_{\text{disc}} \log|P_{\boldsymbol{\kappa}}| \le B_0 + M\Lambda_0 + c_2 + (\sigma_0 - \tfrac12 + 2\varepsilon)\log(T+R) + \max_{\text{disc}} \log|G|.$$

At the centre, by Step 1,

$$\log|P_{\boldsymbol{\kappa}}(\sigma_0+iT)| = \log|Y(\sigma_0+iT)| + \Re\Lambda_{\mathbf{a}}(\sigma_0+iT) + \log|G(\sigma_0+iT)| \ge -B_0 - 2E(T) - M\Lambda_0 + \log|G(\sigma_0+iT)|.$$

Since $|\nabla \log|G(s)|| \le \frac{1}{2}\log\frac{|t|}{2\pi} + c_3$ for $t \ge T_0$, the oscillation of $\log|G|$ over the disc is at most $R(\frac{1}{2}\log\frac{T+R}{2\pi} + c_3)$. Altogether

$$\nu \le \frac{2B_0 + 2M\Lambda_0 + c_4 + 2E(T) + \big(\sigma_0 - \frac{1}{2} + 2\varepsilon + \frac{R}{2}\big)\log(T+R)}{\log(R/r)}. \tag{9}$$

Step 4: the count. $P_{\boldsymbol{\kappa}}$ is entire, real on $\mathbb{R}$, symmetric under $s \mapsto 1-s$, positive at σ_0, with all zeros on $\Re s = \frac{1}{2}$; the argument principle on the rectangle $[1-\sigma_0, \sigma_0] \times [-T, T]$, together with the two symmetries, gives (Titchmarsh, §9.3, applied verbatim)

$$\pi N_{\boldsymbol{\kappa}}(T) = \Delta_{[\sigma_0, \sigma_0+iT]} \arg P_{\boldsymbol{\kappa}} + \Delta_{[\sigma_0+iT, \frac{1}{2}+iT]} \arg P_{\boldsymbol{\kappa}}.$$

Write $P_{\boldsymbol{\kappa}} = G\, e^{\Lambda_{\mathsf{a}}}\, Y$. The two pieces of $\arg G$ give the Riemann–von Mangoldt main term, $\vartheta(T) + \pi + O(1/T) = \pi\mathcal{N}_{\mathrm{RVM}}(T) + O(1/T)$, exactly as for ζ, since G is the same. The vertical piece of $\arg(e^{\Lambda_{\mathsf{a}}}Y)$ is at most $2B_0 + 2E(T) + 2M\Lambda_0$ by (7), since $\log(e^{\Lambda_{\mathsf{a}}}Y) = \log\zeta_{\boldsymbol{\kappa}}$ is a continuous branch on the line. The horizontal piece of $\arg(e^{\Lambda_{\mathsf{a}}}Y)$ equals $\Delta_{\mathrm{seg}}\arg P_{\boldsymbol{\kappa}} - \Delta_{\mathrm{seg}}\arg G$, bounded by $\pi(\nu+1) + c_5$ by Step 3 and Stirling. With $|n(T) - \mathcal{N}_{\mathrm{RVM}}(T)| \le \frac{1}{2}$ this proves (4), with $A_1 = \frac{\sigma_0 - \frac{1}{2} + R/2}{\log(R/r)} + 1$, $A_3 = \frac{4}{\log(R/r)} + 1$ and A_2 collecting the remaining constants.

Step 5: from the count to the displacement. Let $\Phi(T)$ denote the right side of (4), nondecreasing in T. If $\kappa_k > \tau_k + d$ with $d > 0$, then at $T = \tau_k + d$ (perturbed slightly if it is an ordinate) $N_{\boldsymbol{\kappa}}(T) \le k-1$ while $n(T) \ge n(\tau_k) + \lfloor \mathcal{N}_{\mathrm{RVM}}(\tau_k + d) - \mathcal{N}_{\mathrm{RVM}}(\tau_k) \rfloor - 1 \ge k + \frac{d}{2\pi}\log\frac{\tau_k}{2\pi} - 2$, because $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}$ and $\mathcal{N}'_{\mathrm{RVM}}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}$ is increasing; hence $\frac{d}{2\pi}\log\frac{\tau_k}{2\pi} \le \Phi(\tau_k + d) + 1$. The symmetric argument handles $\kappa_k < \tau_k - d$. Since $d \le C$ is known a priori, $\Phi(\tau_k + d) \le \Phi(\tau_k + C)$, and $E(\tau_k + C + R) \le \varepsilon\log(\tau_k + C + R) \le \varepsilon\log\tau_k + \varepsilon\log(1 + C + R)$, which gives (5). The final assertion follows by letting $k \to \infty$. □

Remark 2 (Exact solutions)**.** *If $(\boldsymbol{\kappa}, (\mathsf{a}_n))$ is an exact solution of (1) (no truncation in k or n, masses bounded, $\varepsilon = 0$), the same proof applies without any tube assumption beyond local density $N_{\boldsymbol{\kappa}}(t+1) - N_{\boldsymbol{\kappa}}(t) \ll \log t$, the real-point term being then estimated through $\sum_k |\kappa_k - \tau_k|\tau_k^{-3}$, which converges under local density alone. The conclusion is $|\kappa_k - \tau_k| \le C^* + o(1)$ with C^* absolute. This is Proposition prop:rung-apriori-v18 with the constants made explicit.*

Remark 3 (What a section controls)**.** *A finite section imposes (1) only against its tests. Let the tests be $\{w(u)\,\varphi(u) : \varphi$ band-limited to $|\omega| \le \Omega\}$ with a fixed window $w \in C_c^\infty(0,\infty)$, $w \equiv 1$ on $[u_0, U]$, $\operatorname{supp} w \subset [u_0/2, 2U]$. For $|v| \le \Omega - \Omega_1$ the kernel $w(u)e^{-(\sigma_0 - \frac{1}{2})u}e^{-ivu}$ lies, up to a tail of size $O(\Omega_1^{-A})$, in the span of the tests, so*

$$\widehat{\mathsf{R}}(\sigma_0 + iv) = \underbrace{\langle \mathsf{R},\ w\, e^{-(\sigma_0-\frac{1}{2})u}e^{-ivu}\rangle}_{\text{controlled by the section residual}} + \underbrace{\int_0^{u_0} (1-w)\, e^{-(\sigma_0-\frac{1}{2})u}e^{-ivu}\mathsf{R}(u)\,du}_{\text{near edge}} + \underbrace{\int_U^{\infty} (1-w)\, e^{-(\sigma_0-\frac{1}{2})u}e^{-ivu}\mathsf{R}(u)\,du}_{\text{far edge}}.$$

The far edge is $O(e^{-(\sigma_0-1)U})$ times a polynomial in U, uniformly in v, because R has at most exponential growth $e^{u/2}$ on (U,∞) (the pole term of k) and no atoms there enter the section. The near edge is not small automatically: on $(0, u_0)$ there are no atoms and $\mathsf{R} = \sum_k[\cos(\kappa_k u) - \cos(\tau_k u)] - \mathsf{D}_\tau(u)$, which for a list at distance C from the clock is of size $C\log(1/u)$ as $u \to 0$. Its contribution to $E(T)$ is at most $T\int_0^{u_0}|\mathsf{R}|$, i.e. $O(C\,T\,u_0\log(1/u_0))$, which is $\le \varepsilon_0\log T$ only for $u_0 \le \varepsilon_0\log T/(C\,T\log(1/u_0))$; in a band-limited section u_0 cannot be taken below $1/\Omega$. Therefore the heights T to which Lemma 2 applies with a given ε_0 are

$$T \le T_{\max}(\Omega) := \frac{\varepsilon_0\,\Omega}{C\log\Omega}\,(1 + o(1)),$$

the "interior ordinates" of the section. Equivalently: the quantity that a section must make small is not the Euclidean residual but

$$\sup_{2\le T\le T_{\max}} \frac{1}{\log T}\int_{-T}^{T} \left|\langle \mathsf{R},\ w\, e^{-(\sigma_0-\frac{1}{2})u} e^{-ivu}\rangle\right| dv,$$

and the near edge must be handled either by tests reaching $u_0 \ll 1/T$ or by an independent bound on R near $u = 0$.

Remark 4 (Informative sections)**.** *Where the atoms are denser than the resolution of the tests, $e^{-u} \ll \pi/\Omega$, i.e. $u > \log\Omega + O(1)$, the masses alone synthesize any smooth target by a Riemann sum, $\mathsf{a}_n \approx g(\log n)/n$, with $|\mathsf{a}_n| < 1/e$ and without moving any ordinate. Hence the part of the window above $\log\Omega$ carries no information about the ordinates, and a section informs only if $U \le \log\Omega + O(1)$. In that regime the unknown count is about $e^U(\frac{U}{2\pi}+1)$ against about $\frac{U}{\pi}e^U$ test equations: the system is overdetermined by a factor close to 2, and $E_m \to 0$ is a compatibility statement, not a dimension count. This is why route B is equivalent to (D) and not weaker.*